

Phu Vinh Nguyen

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SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS, Bash, R

Frameworks and Libraries: React, Node.js, Express, Electron, CustomTkinter, Tesseract

Technical Tools: Git, Docker, Jira, LIMS (user level), HPC environments, REST API, SMTP

Database: MySQL, PostgreSQL

WORK EXPERIENCE

Lab Automation Engineer | Application Developer, Sanofi Pasteur

Jun 2025 – Present

- Built a cross-platform RESTful application using Electron, React, and Node.js to automate internal request forms and ticketing systems via SMTP integration, serving 100+ R&D users and reducing on-call workloads by 15%.
- Analyzed and managed testing methods for liquid handler robots to improve automation by 10 times conventional means.
- Oversaw biweekly presentations to primary stakeholders communicating performance, current objectives and future improvements.

Automation Engineer, Foundation Medicine Inc

Nov 2024 – Apr 2025

- Developed a Python GUI application with CustomTkinter that served as a client interface for backend automation systems, enabling 50+ R&D users to execute complex automation workflows independently and reducing support requests by 20%.
- Built a React application that reads and displays Hamilton robot configuration metadata, providing teams with clear visibility into system parameters and object settings reducing troubleshooting time by 40%.
- Created barcode scanning systems and ETL workflows to integrate sample data into Sapio LIMS, automating sample tracking and reducing manual entry errors.

Lab Automation Engineer, Sanofi Pasteur

Oct 2022 – Nov 2024

- Developed a suite of Python scripts to automate routine tasks, including data parsing and data analysis.
- Optimized high-throughput Hamilton liquid handling methods, reducing assay processing times by up to 30% in DNA extraction and fluorescence-based workflows.
- Integrated laboratory automation devices into robotic workflows through TCP/IP and APIs, to enable command execution and data exchange across multiple devices.

Senior Research Associate, Dermbiont Inc.

Oct 2021 – Sept 2022

- Established scalable genomic sequencing workflows using Nanopore to build automated data analysis pipelines with open-source bioinformatics tools to support high-throughput microbiome research.
- Collaborated with external bioinformatics teams to troubleshoot and optimize Dockerized QC pipelines, ensuring consistent data quality standards and reducing processing errors by 25% across research projects.

PROJECTS

AI Hospital Analyzer – Work in progress

- Developing an AI-powered hospital bill analyzer using Gemini API and Tesseract OCR + Ollama to review medical invoices, identifies billing errors and overcharges, and provides detailed cost breakdowns to help patients understand and dispute incorrect charges.

Personal Website

- Developed a React portfolio website to showcase professional experience, technical projects, and programming

EDUCATION

Northeastern University

Certificate in Bioinformatics, concentration in software development

Boston, MA

May 2025

Uniwersytet Jagielloński

Master's in Molecular Biotechnology

Krakow, PL

Jul 2021

University of California, Irvine

Bachelor of Science in Pharmaceutical Sciences

Irvine, CA

Sept 2019